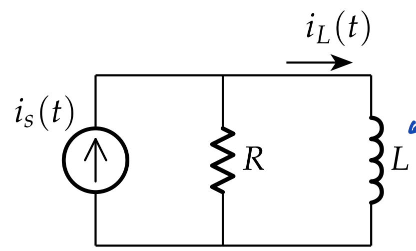
Quiz 9/In-class Assignment

Name: ______________________________ PSID: __________

Problem 1

For the following circuit, $i_s(t) = 12\cos(1000t - 30°)$ [mA], R = 2 [Ω], and L = 1 [mH]. Write a solution for $i_L(t)$.



$Z_L = j\omega L = j\left(1000\left[\frac{rad}{s}\right] \cdot 1[mH]\right) = j[\Omega]$

$$\bar{I}_L = \bar{I}_s \frac{R}{R + Z_L} = 12[mA]\angle{-30°} \cdot \frac{2[\cancel{\Omega}]}{2[\cancel{\Omega}] + j[\cancel{\Omega}]}$$

$$= 12[mA]\angle{-30°} \; \frac{2}{2+j}$$

$$= \left(12[mA]\cos(-30°) + j12[mA]\sin(-30°)\right)\frac{2}{2+j}$$

$$= (5.91 - j8.86)[mA]$$

$$= 10.7[mA]\angle{-56.6°}$$

$$\Rightarrow i_L(t) = 10.7[mA]\cos(1000t - 56.6°)$$

$i_L(t) =$ ______________________

Problem 2

7.39 The voltage source in the circuit of **Fig. P7.39** is given by

$$\upsilon_s(t) = 12\cos 10^4 t \text{ V}.$$

(a) Transform the circuit to the phasor domain and then determine the equivalent impedance **Z** at terminals (a, b). ~~(*Hint:* Application of Δ–Y transformation should prove helpful.)~~

(b) Determine the phasor **I**, corresponding to $i(t)$.

(c) Determine $i(t)$.

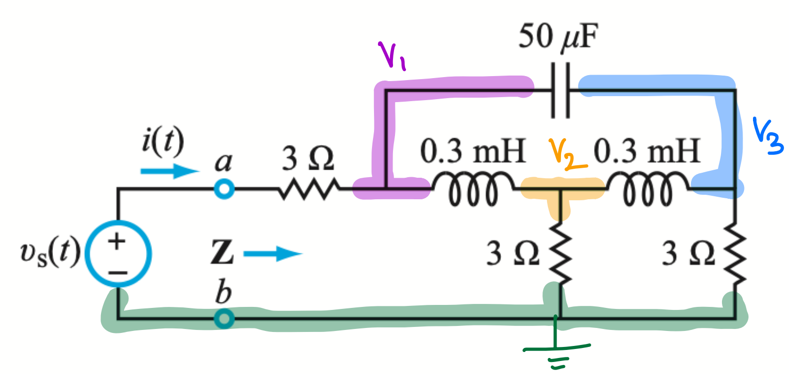


Figure P7.39 Circuit for Problem 7.39.

$Z_1 = j\,10^4[\tfrac{rad}{s}]\,0.3[mH] = j\,10\cdot 0.3[\Omega] = j3[\Omega]$

$Z_2 = \dfrac{-j}{10^4[\tfrac{rad}{s}]\cdot 50[\mu F]} = \dfrac{-j}{50\cdot 10^{-2}}[\Omega] = -j2[\Omega]$

$\left[\dfrac{\bar{V}_1 - 12[V]}{3[\Omega]} + \dfrac{\bar{V}_1 - \bar{V}_2}{j3[\Omega]} + \dfrac{\bar{V}_1 - \bar{V}_3}{-j2[\Omega]} = 0\right] \times 6[\Omega]$

$2\bar{V}_1 - 24[V] - j2(\bar{V}_1 - \bar{V}_2) + j3(\bar{V}_1 - \bar{V}_3) = 0$

$(2+j)\bar{V}_1 + j2\bar{V}_2 - j3\bar{V}_3 = 24[V]$

$\left[\dfrac{\bar{V}_2 - \bar{V}_1}{j3[\Omega]} + \dfrac{\bar{V}_2}{3[\Omega]} + \dfrac{\bar{V}_2 - \bar{V}_3}{j3[\Omega]} = 0\right] \times 3[\Omega]$

$-j\bar{V}_2 + j\bar{V}_1 + \bar{V}_2 - j\bar{V}_2 + j\bar{V}_3 = 0$

$j\bar{V}_1 + (1-j2)\bar{V}_2 + j\bar{V}_3 = 0$

$\times 6[\Omega]\left[\dfrac{\bar{V}_3 - \bar{V}_1}{-j2[\Omega]} + \dfrac{\bar{V}_3 - \bar{V}_2}{j3[\Omega]} + \dfrac{\bar{V}_3}{3[\Omega]} = 0\right]$

$j3\bar{V}_3 - j3\bar{V}_1 - j2\bar{V}_3 + j2\bar{V}_2 + 2\bar{V}_3 = 0$

$-j3\bar{V}_1 + j2\bar{V}_2 + (2+j)\bar{V}_3 = 0$

Using calculator:

$\bar{V}_1 = \left(\dfrac{27}{5} + j\dfrac{9}{5}\right)[V] = (5.4 - j1.8)[V]$

$\bar{V}_2 = j\dfrac{12}{5}[V] = j2.4[V]$

$\bar{V}_3 = \left(3 + j\dfrac{33}{5}\right)[V] = (3 + j6.6)[V]$

$$\bar{I} = \frac{12[V] - (5.4 - j1.8)[V]}{3[\Omega]} = (2.2 + j0.6)[A]$$

$$= 2.28[A]\angle 15.2^\circ$$

$$Z = \frac{\bar{V}}{\bar{I}} = \frac{12[V]}{(2.2+j0.6)[A]} = (5.08 - j1.38)[\Omega]$$

$$i(t) = 2.28[A]\cos\left(10^4\left[\frac{rad}{s}\right]t + 15.2^\circ\right)$$